

Part I — Limit Process / Definition of the Derivative

KEY DEFINITION (FIRST PRINCIPLES)

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

This is the *definition of the derivative*. Every answer in Questions 1–4 must use this limit.

Question 1 | $f(x) = x^2 - 5x + 3$

STEP 1 — WRITE F(X+H)

$$f(x+h) = (x+h)^2 - 5(x+h) + 3 = x^2 + 2xh + h^2 - 5x - 5h + 3$$

STEP 2 — COMPUTE F(X+H) - F(X)

$$\begin{aligned} f(x+h) - f(x) &= (x^2 + 2xh + h^2 - 5x - 5h + 3) - (x^2 - 5x + 3) \\ &= 2xh + h^2 - 5h \end{aligned}$$

STEP 3 — DIVIDE BY H

$$\frac{f(x+h) - f(x)}{h} = \frac{h(2x + h - 5)}{h} = 2x + h - 5$$

STEP 4 — TAKE THE LIMIT AS H → 0

$$f'(x) = \lim_{h \rightarrow 0} (2x + h - 5) = 2x - 5$$

✓ **ANSWER:** $f'(x) = 2x - 5$

Question 2 | $f(x) = \frac{1}{x-1}$

STEP 1 — WRITE F(X+H)

$$f(x+h) = \frac{1}{(x+h)-1} = \frac{1}{x+h-1}$$

STEP 2 — COMPUTE F(X+H) - F(X)

$$\begin{aligned} f(x+h) - f(x) &= \frac{1}{x+h-1} - \frac{1}{x-1} \\ &= \frac{(x-1) - (x+h-1)}{(x+h-1)(x-1)} \\ &= \frac{-h}{(x+h-1)(x-1)} \end{aligned}$$

STEP 3 — DIVIDE BY H

$$\frac{f(x+h) - f(x)}{h} = \frac{-h}{h(x+h-1)(x-1)} = \frac{-1}{(x+h-1)(x-1)}$$

STEP 4 – TAKE THE LIMIT AS $h \rightarrow 0$

$$f'(x) = \lim_{h \rightarrow 0} \frac{-1}{(x+h-1)(x-1)} = \frac{-1}{(x-1)^2}$$

✓ **ANSWER:** $f'(x) = \frac{-1}{(x-1)^2}$

Question 3 | $f(x) = \sqrt{x-2}$

STEP 1 – WRITE $f(x+h)$

$$f(x+h) = \sqrt{x+h-2}$$

STEP 2 – FORM THE DIFFERENCE QUOTIENT

$$\frac{f(x+h) - f(x)}{h} = \frac{\sqrt{x+h-2} - \sqrt{x-2}}{h}$$

STEP 3 – RATIONALISE THE NUMERATOR (MULTIPLY BY CONJUGATE)

$$\begin{aligned} &= \frac{\sqrt{x+h-2} - \sqrt{x-2}}{h} \cdot \frac{\sqrt{x+h-2} + \sqrt{x-2}}{\sqrt{x+h-2} + \sqrt{x-2}} \\ &= \frac{(x+h-2) - (x-2)}{h(\sqrt{x+h-2} + \sqrt{x-2})} \\ &= \frac{h}{h(\sqrt{x+h-2} + \sqrt{x-2})} \\ &= \frac{1}{\sqrt{x+h-2} + \sqrt{x-2}} \end{aligned}$$

STEP 4 – TAKE THE LIMIT AS $h \rightarrow 0$

$$f'(x) = \lim_{h \rightarrow 0} \frac{1}{\sqrt{x+h-2} + \sqrt{x-2}} = \frac{1}{2\sqrt{x-2}}$$

✓ **ANSWER:** $f'(x) = \frac{1}{2\sqrt{x-2}}$

Question 4 | Prove from first principles that $\frac{d}{dx}[\cos x] = -\sin x$

TRIGONOMETRIC IDENTITIES REQUIRED

$\cos(A+B) = \cos A \cos B - \sin A \sin B$ and the standard limits: $\lim_{h \rightarrow 0} \frac{\sin h}{h} = 1$, $\lim_{h \rightarrow 0} \frac{\cos h - 1}{h} = 0$

STEP 1 – APPLY THE DEFINITION

$$\frac{d}{dx}[\cos x] = \lim_{h \rightarrow 0} \frac{\cos(x+h) - \cos x}{h}$$

STEP 2 – EXPAND $\cos(x+h)$ USING THE ADDITION FORMULA

$$= \lim_{h \rightarrow 0} \frac{\cos x \cos h - \sin x \sin h - \cos x}{h}$$

STEP 3 — GROUP AND FACTOR

$$= \lim_{h \rightarrow 0} \left[\cos x \cdot \frac{\cos h - 1}{h} - \sin x \cdot \frac{\sin h}{h} \right]$$

STEP 4 — APPLY THE STANDARD LIMITS

$$= \cos x \cdot (0) - \sin x \cdot (1) = -\sin x$$

✓ **PROVED:** $\frac{d}{dx}[\cos x] = -\sin x$ ■

Part II — Basic Rules of Differentiation

RULES USED IN QUESTIONS 5–7

• **POWER RULE:**

$$\frac{d}{dx}[x^n] = nx^{n-1}$$

• **CONSTANT RULE:**

$$\frac{d}{dx}[c] = 0$$

• **CONSTANT MULTIPLE RULE:**

$$\frac{d}{dx}[cf(x)] = cf'(x)$$

• **SUM/DIFFERENCE RULE:**

$$\frac{d}{dx}[f \pm g] = f' \pm g'$$

• **EXPONENTIAL:**

$$\frac{d}{dx}[e^x] = e^x$$

• **LOGARITHM:**

$$\frac{d}{dx}[\ln x] = \frac{1}{x}$$

• **TRIG:**

$$\frac{d}{dx}[\sin x] = \cos x$$

Question 5 | $f(x) = \frac{1}{4}x^4 + 5x^3 + 20 - 2x + 3x^{-3} + e^x$

Differentiate term by term using the Power Rule, Constant Rule, and $\frac{d}{dx}[e^x] = e^x$:

$$\begin{aligned} f'(x) &= \frac{1}{4}(4x^3) + 5(3x^2) + 0 - 2(1) + 3(-3x^{-4}) + e^x \\ &= x^3 + 15x^2 - 2 - 9x^{-4} + e^x \end{aligned}$$

✓ **ANSWER:** $f'(x) = x^3 + 15x^2 - 2 - 9x^{-4} + e^x$

Question 6 | $f(x) = \frac{3}{x^2} - \ln x + 6\sqrt{x} + 5 \sin x$

STEP 1 — REWRITE USING EXPONENTS

$$f(x) = 3x^{-2} - \ln x + 6x^{1/2} + 5 \sin x$$

STEP 2 — DIFFERENTIATE TERM BY TERM

$$\begin{aligned} f'(x) &= 3(-2x^{-3}) - \frac{1}{x} + 6\left(\frac{1}{2}x^{-1/2}\right) + 5 \cos x \\ &= -6x^{-3} - \frac{1}{x} + 3x^{-1/2} + 5 \cos x \end{aligned}$$

✓ **ANSWER:** $f'(x) = -\frac{6}{x^3} - \frac{1}{x} + \frac{3}{\sqrt{x}} + 5 \cos x$

Question 7 | $h(x) = \frac{4x^3 + 2x + 5}{2x}$

STEP 1 – SPLIT THE FRACTION (SIMPLIFY FIRST)

$$h(x) = \frac{4x^3}{2x} + \frac{2x}{2x} + \frac{5}{2x} = 2x^2 + 1 + \frac{5}{2}x^{-1}$$

STEP 2 – DIFFERENTIATE TERM BY TERM

$$h'(x) = 4x + 0 + \frac{5}{2}(-1)x^{-2} = 4x - \frac{5}{2x^2}$$

✓ **ANSWER:** $h'(x) = 4x - \frac{5}{2x^2}$

Part III – Rates of Change

Question 8 | **Tangent line to $f(x) = x^3 - 2x$ at $x = -1$**

TANGENT LINE FORMULA

$$y - y_1 = m(x - x_1), \text{ where } m = f'(x_1)$$

STEP 1 – FIND THE Y-COORDINATE

$$f(-1) = (-1)^3 - 2(-1) = -1 + 2 = 1 \quad \Rightarrow \quad \text{Point: } (-1, 1)$$

STEP 2 – FIND THE DERIVATIVE (SLOPE FUNCTION)

$$f'(x) = 3x^2 - 2$$

STEP 3 – EVALUATE SLOPE AT X = -1

$$m = f'(-1) = 3(-1)^2 - 2 = 3 - 2 = 1$$

STEP 4 – WRITE THE TANGENT LINE EQUATION

$$y - 1 = 1(x - (-1)) \Rightarrow y = x + 2$$

✓ **ANSWER:** $y = x + 2$

Question 9 | **Tangent line given $f(1) = -3$ and $f'(1) = 6$**

The point of tangency is $(1, -3)$ and the slope is $m = f'(1) = 6$.

$$y - (-3) = 6(x - 1) \Rightarrow y + 3 = 6x - 6 \Rightarrow y = 6x - 9$$

✓ **ANSWER:** $y = 6x - 9$

Question 10 | **Ball thrown upward: $s(t) = -16t^2 + 64t$**

Part (a)

The velocity function is the derivative of position:

$$v(t) = s'(t) = \frac{d}{dt}(-16t^2 + 64t) = -32t + 64$$

✓ ANSWER (A): $v(t) = -32t + 64$ m/s

Part (b)

Maximum height occurs when $v(t) = 0$ (velocity changes sign from positive to negative):

$$-32t + 64 = 0 \implies t = 2 \text{ s}$$

✓ ANSWER (B): $t = 2$ seconds

Part (c)

The velocity at maximum height is, by definition:

$$v(2) = -32(2) + 64 = -64 + 64 = 0 \text{ m/s}$$

✓ ANSWER (C): $v = 0$ m/s (the ball is momentarily at rest at the peak)

Part IV — Product and Quotient Rules

PRODUCT RULE

$$\frac{d}{dx}[uv] = u'v + uv'$$

QUOTIENT RULE

$$\frac{d}{dx}\left[\frac{u}{v}\right] = \frac{u'v - uv'}{v^2}$$

Question 11 | $y = 6x^2 \cos x$

Let $u = 6x^2 \Rightarrow u' = 12x$ and $v = \cos x \Rightarrow v' = -\sin x$.

$$\frac{dy}{dx} = u'v + uv' = 12x \cos x + 6x^2(-\sin x)$$

✓ ANSWER: $\frac{dy}{dx} = 12x \cos x - 6x^2 \sin x$

Question 12 | $y = (3x^2 - 4\sqrt{x})(x^3 + 5)$

Let $u = 3x^2 - 4x^{1/2} \Rightarrow u' = 6x - 2x^{-1/2}$ and $v = x^3 + 5 \Rightarrow v' = 3x^2$.

$$\begin{aligned} \frac{dy}{dx} &= (6x - 2x^{-1/2})(x^3 + 5) + (3x^2 - 4x^{1/2})(3x^2) \\ &= 6x^4 + 30x - 2x^{5/2} - 10x^{-1/2} + 9x^4 - 12x^{5/2} \\ &= 15x^4 + 30x - 14x^{5/2} - 10x^{-1/2} \end{aligned}$$

✓ ANSWER: $\frac{dy}{dx} = 15x^4 - 14x^{5/2} + 30x - \frac{10}{\sqrt{x}}$

Question 13 | $h(t) = \left(\frac{t^2}{5} - t^2\right) \sec t$

Note: The expression $\frac{t^2}{5} - t^2 = \frac{t^2 - 5t^2}{5} = -\frac{4t^2}{5}$. We apply the Product Rule.

Let $u = -\frac{4}{5}t^2 \Rightarrow u' = -\frac{8}{5}t$ and $v = \sec t \Rightarrow v' = \sec t \tan t$.

$$\begin{aligned} h'(t) &= -\frac{8}{5}t \sec t + \left(-\frac{4}{5}t^2\right) \sec t \tan t \\ &= -\frac{4}{5}t \sec t (2 + t \tan t) \end{aligned}$$

✓ ANSWER: $h'(t) = -\frac{8}{5}t \sec t - \frac{4}{5}t^2 \sec t \tan t$

Question 14 | $g(x) = \frac{x^2}{2\sqrt{x} + 1}$

Use the Quotient Rule: $u = x^2$, $u' = 2x$, $v = 2\sqrt{x} + 1 = 2x^{1/2} + 1$, $v' = x^{-1/2}$.

$$\begin{aligned} g'(x) &= \frac{2x(2\sqrt{x} + 1) - x^2 \cdot x^{-1/2}}{(2\sqrt{x} + 1)^2} \\ &= \frac{4x^{3/2} + 2x - x^{3/2}}{(2\sqrt{x} + 1)^2} \\ &= \frac{3x^{3/2} + 2x}{(2\sqrt{x} + 1)^2} \end{aligned}$$

✓ ANSWER: $g'(x) = \frac{3x\sqrt{x} + 2x}{(2\sqrt{x} + 1)^2}$

Question 15 | $f(t) = \frac{3 - \frac{1}{t}}{t + 1}$

STEP 1 – SIMPLIFY NUMERATOR

$$f(t) = \frac{\frac{3t-1}{t}}{t+1} = \frac{3t-1}{t(t+1)}$$

STEP 2 – APPLY QUOTIENT RULE WITH $U = 3T-1$, $V = T^2+T$

$$\begin{aligned} u' &= 3, \quad v' = 2t + 1 \\ f'(t) &= \frac{3(t^2 + t) - (3t - 1)(2t + 1)}{(t^2 + t)^2} \\ &= \frac{3t^2 + 3t - (6t^2 + 3t - 2t - 1)}{t^2(t + 1)^2} \\ &= \frac{3t^2 + 3t - 6t^2 - 3t + 2t + 1}{t^2(t + 1)^2} \\ &= \frac{-3t^2 + 2t + 1}{t^2(t + 1)^2} \end{aligned}$$

✓ ANSWER: $f'(t) = \frac{-3t^2 + 2t + 1}{t^2(t + 1)^2}$

Question 16 | Prove by Quotient Rule that $\frac{d}{dx}[\tan x] = \sec^2 x$

IDENTITY NEEDED:

$\sin^2 x + \cos^2 x = 1$, and $\sec x = \frac{1}{\cos x}$

Write $\tan x = \frac{\sin x}{\cos x}$. Let $u = \sin x$, $u' = \cos x$, $v = \cos x$, $v' = -\sin x$.

$$\begin{aligned}\frac{d}{dx}[\tan x] &= \frac{\cos x \cdot \cos x - \sin x \cdot (-\sin x)}{\cos^2 x} \\ &= \frac{\cos^2 x + \sin^2 x}{\cos^2 x} \\ &= \frac{1}{\cos^2 x} = \sec^2 x\end{aligned}$$

✓ **PROVED:** $\frac{d}{dx}[\tan x] = \sec^2 x$ ■

Part V — Higher Order Derivatives

NOTATION:

$f''(x) = \frac{d^2}{dx^2}[f]$, $f'''(x) = \frac{d^3}{dx^3}[f]$, $f^{(n)}(x) = \frac{d^n}{dx^n}[f]$. Apply the Power Rule repeatedly.

Question 17 | $f(x) = x^2 - \cos x$, find $f'''(x)$

$$\begin{aligned}f(x) &= x^2 - \cos x \\ f'(x) &= 2x + \sin x \\ f''(x) &= 2 + \cos x \\ f'''(x) &= -\sin x\end{aligned}$$

✓ **ANSWER:** $f'''(x) = -\sin x$

Question 18 | Given $f'(x) = \frac{1}{2}x^4 - \frac{2}{x}$, find $f^{(4)}(x)$

REWRITE

$$f'(x) = \frac{1}{2}x^4 - 2x^{-1}$$

$$\begin{aligned}f''(x) &= 2x^3 + 2x^{-2} \\ f'''(x) &= 6x^2 - 4x^{-3} \\ f^{(4)}(x) &= 12x + 12x^{-4}\end{aligned}$$

✓ **ANSWER:** $f^{(4)}(x) = 12x + \frac{12}{x^4}$

Question 19 | Given $f^{(4)}(x) = 2\sqrt{x}$, find $f^{(5)}(x)$

Rewrite: $f^{(4)}(x) = 2x^{1/2}$. Differentiate once more:

$$f^{(5)}(x) = 2 \cdot \frac{1}{2}x^{-1/2} = x^{-1/2} = \frac{1}{\sqrt{x}}$$

✓ **ANSWER:** $f^{(5)}(x) = \frac{1}{\sqrt{x}}$

Part VI — Chain Rule

CHAIN RULE:

If $y = f(g(x))$, then $\frac{dy}{dx} = f'(g(x)) \cdot g'(x)$

Equivalently: differentiate the *outer* function, keep the *inner* function, then multiply by the derivative of the *inner* function.

Question 20 | $g(x) = 3(4 - 9x)^4$

Outer: $(\cdot)^4$, Inner: $u = 4 - 9x \Rightarrow u' = -9$

$$g'(x) = 3 \cdot 4(4 - 9x)^3 \cdot (-9) = -108(4 - 9x)^3$$

✓ ANSWER: $g'(x) = -108(4 - 9x)^3$

Question 21 | $f(x) = \sqrt[3]{12x - 5 \cos x}$

REWRITE

$$f(x) = (12x - 5 \cos x)^{1/3}$$

Outer: $(\cdot)^{1/3}$, Inner: $u = 12x - 5 \cos x \Rightarrow u' = 12 + 5 \sin x$

$$f'(x) = \frac{1}{3}(12x - 5 \cos x)^{-2/3} \cdot (12 + 5 \sin x)$$

✓ ANSWER: $f'(x) = \frac{12 + 5 \sin x}{3(12x - 5 \cos x)^{2/3}}$

Question 22 | $f(t) = \frac{-3}{(t - 2)^4}$

REWRITE

$$f(t) = -3(t - 2)^{-4}$$

Outer: $(\cdot)^{-4}$, Inner: $u = t - 2 \Rightarrow u' = 1$

$$f'(t) = -3 \cdot (-4)(t - 2)^{-5} \cdot 1 = \frac{12}{(t - 2)^5}$$

✓ ANSWER: $f'(t) = \frac{12}{(t - 2)^5}$

Question 23 | $f(x) = \sin(2x + 3)$

Outer: $\sin(\cdot)$, Inner: $u = 2x + 3 \Rightarrow u' = 2$

$$f'(x) = \cos(2x + 3) \cdot 2 = 2 \cos(2x + 3)$$

✓ ANSWER: $f'(x) = 2 \cos(2x + 3)$

Question 24 | $f(x) = -5 \cos(\pi x^2)$

Outer: $\cos(\cdot)$, Inner: $u = \pi x^2 \Rightarrow u' = 2\pi x$

$$f'(x) = -5 \cdot (-\sin(\pi x^2)) \cdot 2\pi x = 10\pi x \sin(\pi x^2)$$

✓ **ANSWER:** $f'(x) = 10\pi x \sin(\pi x^2)$

Question 25 | $f(\theta) = \sin^2(3\theta)$

TWO-LAYER CHAIN RULE:

rewrite as $[\sin(3\theta)]^2$. Differentiate the outer power, then $\sin(3\theta)$, then the innermost 3θ .

Let $w = \sin(3\theta)$, so $f = w^2$.

$$f'(\theta) = 2w \cdot \frac{dw}{d\theta} = 2 \sin(3\theta) \cdot \cos(3\theta) \cdot 3 = 6 \sin(3\theta) \cos(3\theta)$$

OPTIONAL SIMPLIFICATION USING DOUBLE-ANGLE IDENTITY $\sin 2A = 2 \sin A \cos A$

$$f'(\theta) = 3 \sin(6\theta)$$

✓ **ANSWER:** $f'(\theta) = 6 \sin(3\theta) \cos(3\theta) = 3 \sin(6\theta)$